

# Advancements and Future Perspectives in Nutrient Film Technique Hydroponic System: A Comprehensive Review and Bibliometric Analysis

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## Abstract

In the context of climate change, reducing the environmental impact of agriculture has become increasingly critical. The Nutrient Film Technique (NFT) is a hydroponic system designed to optimize water and nutrient use. This bibliometric review examines the evolution of NFT research from 1977 to 2023, analyzing 774 scientific documents. The review highlights an upward trend in NFT-related studies with a shift from conference papers to peer-reviewed journal articles. Lettuce and tomatoes are primary crops studied. Research on lettuce focused on nitrate accumulation and biofortification; tomato studies explored oxygen concentration challenges. Innovations such as the Nutrient Drip Technique (NDT) and New Growing System (NGS) show promise. Integration of NFT with aquaponics highlights potential for sustainable, water-efficient crop production.

**Keywords:** water use efficiency · soilless · nutrient solution management · climate changes · environmental impact · hydroponics

## 1. Introduction

Climate change is significantly affecting agricultural activities by altering weather patterns, increasing extreme events, and shifting growing seasons. Traditional farming is particularly vulnerable, relying on stable environmental conditions. Innovative practices such as greenhouse farming and soilless cultivation (hydroponics and aeroponics) offer promising solutions. Soilless culture is defined as any method of growing plants without soil, where inorganic nutrients are supplied via irrigation water. The NFT system was developed at the Glasshouse Crops Research Institute (Littlehampton, England) in the late 1960s.

NFT offers several advantages: it eliminates soil-related issues such as erosion and compaction; uses significantly less water than traditional farming; reduces the risk of soil-borne diseases and pests; and is space-efficient, making it suitable for urban farming. Plants in NFT tend to grow faster due to direct nutrient delivery and optimal root oxygenation within the nutrient film.

However, NFT also has challenges. It is most effective for short-term crops with 30-50 day growth cycles. The limited volume of nutrient solution (NS) is susceptible to temperature fluctuations, the system has limited buffering capacity against supply interruptions, and close proximity of plant roots increases disease transmission risk.

## 2. Materials and Methods

A comprehensive search was conducted in April 2024 using Scopus and ISI Web of Science. After removing 162 duplicate articles and 283 irrelevant articles after manual review, a final dataset of 774 records (1977-2023) was analyzed using Bibliometrix (R package/Biblioshiny interface) and Orange data mining software.

### 3. Results

#### 3.1 Publication Trends (1977-2023)

The first scientific work on NFT was published in 1977. Production reached a peak of 81 papers in 2023, with steady increase from 2013. NFT publications grew from 1% of 'soilless' publications in the 1990s to approximately 5% in 2023, reflecting growing relevance. Open Access papers rose from 19% (1977-2010) to 56% (2011-2023).

#### 3.2 Leading Journals and Authors

Acta Horticulturae leads with 94 NFT papers. Most prolific authors: Mortley (25 papers, 1991-2017), Bonsi and Ghey (22 each), Hill (19), Kowalska (16). Brazil and the USA each lead national output with 106 publications, followed by Japan, Italy, India, Poland, Egypt, and Greece.

#### 3.3 Key Topics and Crops

'Hydroponics' appears in 26% of publications; 'NFT' in 24%; 'lettuce' in 11%; 'tomato' 6%; 'salinity' 5%; 'greenhouse' and 'aquaponics' 3% each.

#### 3.4 Key Issues and Challenges

##### 3.4.1 Salinity Tolerance

Salinity stress is a significant NFT constraint as salinity builds up in recirculating NS. Mitigation strategies: selection of salt-tolerant cultivars; grafting salt-sensitive cultivars onto salt-tolerant rootstocks; adjusting NS potassium levels; applying biostimulants such as glycinebetaine; advanced monitoring and control systems.

##### 3.4.2 Nutrient Management

Constant monitoring and adjustment is required to address varying nutrient requirements by species/growth stage, nutrient interactions (e.g., high potassium inhibiting magnesium uptake), and nutrient losses through plant uptake, volatilization, and precipitation. Sensor-based monitoring and precision fertigation are emerging solutions.

##### 3.4.3 Oxygen Availability

Adequate dissolved oxygen (>5 mg/L) is essential. Oxygen fluctuates due to temperature, plant respiration, and microbial activity. High plant density, elevated temperatures (reducing O<sub>2</sub> solubility), and organic matter accumulation all contribute to oxygen stress.

##### 3.4.4 Climate Change and Sustainability

Water scarcity, heat stress, and increased pest pressure are growing challenges. NFT energy consumption (pumps, aeration) and salinity/nitrate accumulation in recirculating NS require ongoing innovation. Renewable energy could significantly reduce the carbon footprint of NFT operations.

#### 3.5 Opportunities

##### 3.5.1 Vertical Farming Integration

NFT is well-suited for vertical farming: higher yields from precise environmental control; hydroponic lettuce uses ~10% of the water of open-field cultivation; year-round production; reduced

transportation costs near urban centers.

### 3.5.2 Biofortification

Precise nutrient delivery makes NFT ideal for biofortification. Studies demonstrate: iodine biofortification (glycinebetaine at 10 mM reduces nitrate by 29%, increases amino acids); selenium biofortification; iron biofortification in lettuce. Salicylic acid in NS influences iodine/selenium accumulation.

### 3.5.3 Integration with Aquaponics

Aquaponic-NFT systems combine fish production with plant cultivation. Fish waste provides nutrients; plants purify water. Optimum ratio: 15-42 g fish feed per m<sup>2</sup> growing area. Benefits include enhanced nutrient utilization, improved water quality, increased production, and elimination of synthetic fertilizers. Common fish species: Nile Tilapia, Empurau, Hoven Carp, Silver Barb.

## 3.6 Lettuce in NFT

Lettuce appears in 107 papers. Key research: nitrate accumulation (reduced by LED red/blue spectra; suspending fertilizer 2-4 days pre-harvest; reducing NO<sub>3</sub>:NH<sub>4</sub><sup>+</sup> ratio; glycinebetaine up to 10 mM) and biofortification with iodine, selenium, and iron. NFT's substrate-free design is ideal for studying nutrient absorption without substrate interference.

## 3.7 Tomato in NFT

Tomato appears in 54 documents. Primary challenge: dissolved oxygen in NS (below 5 mg/L causes stress). Innovations: Nutrient Drip Technique (NDT) with multiple drip points; New Growing System (NGS) with 5-layer design reducing oxygen drop from 6.2 to 2.9 mg/L (NFT) vs 7.12 to 6.65 mg/L (NGS) over 20 m. Note: In commercial settings, NFT is primarily used for short-cycle crops like lettuce, not long-cycle tomatoes.

## 4. Conclusion

This bibliometric review of 774 NFT documents (1977-2023) reveals growing publication trends, a shift from conference papers to peer-reviewed articles, and increasing research on sustainable applications. Lettuce and tomato are primary crops; key challenges include salinity tolerance, nutrient management, oxygen availability, and climate adaptation. NFT integration with aquaponics and vertical farming presents significant opportunities. Continued innovation in nutrient delivery, environmental control, and new crop varieties is essential.

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